Total HW: 2.18, 22, 29, 30

Problem 2.18. Use Stirling's approximation to show that the multiplicity of an Einstein solid, for any large values of N and q, is approximately

$$\Omega(N,q) \approx \frac{\left(\frac{q+N}{q}\right)^q \left(\frac{q+N}{N}\right)^N}{\sqrt{2\pi q(q+N)/N}}.$$

The square root in the denominator is merely large, and can often be neglected. However, it is needed in Problem 2.22. (Hint: First show that $\Omega = \frac{N}{q+N}\frac{(q+N)!}{q!\,N!}$. Do not neglect the $\sqrt{2\pi N}$ in Stirling's approximation.)

$$\Omega(N,q) = \frac{(q+N-1)!}{q!\,(N-1)!}$$

$$\Omega(N,q) = \frac{N}{q+N}\cdot\frac{(q+N)!}{q!\,N!}$$

Now, apply Stirling's approx

$$N! = N\cdot\underbrace{(N-1)\cdot(N-2)\cdot\ldots\cdot 1}_{(N-1)!}$$

$$N! = N\cdot(N-1)!$$

$$(N-1)! = \frac{N!}{N}$$

$$(q+N-1)! = \frac{(q+N)!}{(q+N)}$$

$$N! = \left(\frac{N}{e}\right)^N\cdot\sqrt{2\pi N} = N^N e^{-N}\cdot\sqrt{2\pi N}$$

test Stirling's w/ python

$$\Omega = \frac{N}{q+N} \frac{\left(\frac{q+N}{e}\right)^{q+N} \sqrt{2\pi (q+N)}}{\left(\frac{q}{e}\right)^{q} \sqrt{2\pi q} \left(\frac{N}{e}\right)^{N} \sqrt{2\pi N}}$$

$$= \frac{(N)^{2/2}}{(q+N)^{2/2}} \frac{(q+N)^{q} (q+N)^{N} (q+N)^{1/2}}{q^{q} \cdot N^{N} (2\pi q N)^{1/2}}$$

$$= \left(\frac{q+N}{q}\right)^{q} \left(\frac{q+N}{N}\right)$$